Computer Netzwerke und verteilte Systeme Summer 2026

Exercise 6: Transport

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Deadline: The deadline for this exercise is **03.06.2026, 23:55** and in total **23 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.
For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.
Refrain from copying and pasting your answers, as this is considered `plagiarism` and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!
Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting **(ensure that your handwriting is legible)**.
Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Alternating Bit Protocol 1

Assume a satellite connection with a data rate of $40\frac{\text{Gibit}}{s}$ and a propagation delay of $67\ \text{ms}$. Packets sent over the connection have a size of $32\ \text{KiB}$ ($\text{Ki} = 2^{10}, \text{Mi} = 2^{20}, \ldots$)

(a) How many seconds does it take to transfer $5.1\ \text{MiB}$ over a satellite connection using the alternating bit protocol?

Hint: *Assume that the transmission time of an ACK is equal to the transmission time of a packet. Provide your result with four decimal places.*

Hint: *Converting your results into smaller units saves space and increases readability.*

(b) How many seconds does it take to transfer $5.1\ \text{MiB}$ over the satellite connection in case only the whole transfer is acknowledged at the end? Provide your result with four decimal places.

Task 2: Group Exercise: Transport Layer Theory 1

(a) What is the difference of logical communication between the Transport Layer and Network Layer?

(b) Explain the differences between TCP and UDP regarding connection type, reliability, overhead, and use cases.

(c) Explain the Two Army Problem. Does the Three-Way Handshake solve it?

Task 3: Group Exercise: Transport Layer Theory 2

(a) Explain the difference between Host-to-Host and End-to-End communication.

(b) How does demultiplexing work on the transport layer of the Internet?

(c) The service primitive of the transport layer for the connection release phase is T-Disconnect, which includes T-Disconnect.req(userdata) and T-Disconnect.ind(cause, userdata). Why do we only need the parameter cause for T-Disconnect.ind, but not for T-Disconnect.req?

Hint: Consider two scenarios - one is where the connection teardown is initiated by a service user, and the other one is where the connection teardown is caused by the service provider or due to the network itself.

(d) What is the purpose of sequence numbers? What are the problems related to the use of sequence numbers? List two possible answers for each question.

Task 4: Transport Layer Theory 3 (8 Points)

(a) What are the Core Protocol Functions of the Transport Layer? Name 2. (1 P)

(b) Given the following socket pair $(130.83.47.181, 443)$. Explain what the individual components of this socket pair represent. What do they stand for? (2 P)

Hint: You can use a reverse DNS lookup tool to identify the host behind the IP address.

(c) The Three-Way Handshake alone is not sufficient for establishing a connection. Name and explain the specific problem that arises. Also name and explain a solution to this problem. (3 P)

(d) What is the major limitation of Alternating Bit Protocol regarding network efficiency? Referring to the lecture material, explicitly state under what network conditions this limitation becomes severe. (2 P)

Hint: Recall the mathematical performance analysis discussed in the lecture (e.g., the communication scenario between Frankfurt and New York).

Attention: Submissions without references will not be graded!

Task 5: Ports and Operating Systems (9 Points)

(a) The Internet uses $(\text{IP address}, \text{Port number})$ pairs to identify communication endpoints. Explain why this mechanism violates the layer abstraction of the OSI model. Despite this violation, the mechanism is still used in practice. Give two advantages of using sockets over strict OSI-compliant addressing. (2.5 P)

(b) In the lecture, we discussed why the operating system does not simply use its own Process ID (PID) to address processes on the transport layer. Name two reasons why using PIDs would be problematic, and what is used instead. (2 P)

(c) Consider the following scenario: Two different clients on different hosts send requests to the **same server port** on the same server machine. How many sockets are needed on the server side if both requests use UDP? How many if both use TCP? Explain why. (2.5 P)

(d) Research the default port ranges for **registered** and **ephemeral** ports supported by modern windows versions. (e.g. windows 10 / 11) (2 P)

Attention: Submissions without references will not be graded!

Task 6: Alternating Bit Protocol 2 (6 Points)

Assume a satellite connection with a data rate of $35\frac{\text{Gibit}}{s}$ and a propagation delay of $84\ \text{ms}$. Packets sent over the connection have a size of $13\ \text{KiB}$ $(\text{Ki} = 2^{10}, \text{Mi} = 2^{20}, \ldots)$

(a) How many seconds does it take to transfer 3.7 MiB over a satellite connection using the alternating bit protocol? (4 P)

Hint: *Assume that the transmission time of an ACK is equal to the transmission time of a packet. Provide your result with four decimal places.*

Hint: *Converting your results into smaller units saves space and increases readability.*

(b) How many seconds does it take to transfer 3.7 MiB over the satellite connection in case only the whole transfer is acknowledged at the end? Provide your result with four decimal places. (2 P)